KRRISH GUPTA

Software Engineer Intern | TypeScript ● React ● Node.js ● Express ● SQL ● REST APIs ● LLMs & Vector Databases

Delhi, India | +91-83075-13569 | Krrishg1977@gmail.com | github.com/krrish5235 | linkedin.com/in/krrish-gupta-a34492272

PROFESSIONAL SUMMARY

Final-year B.Tech Computer Science student (2027) and full-stack builder who creates greenfield AI products from 0 to 1, taking ideas from whiteboard to working prototype to deployment: React/TypeScript and Node.js/Express applications on SQL and MongoDB databases, REST APIs, and AI products that wire LLMs, embeddings and vector databases into real workflows. Two remote internships in agile teams, four live-deployed projects, and an ownership-first mindset that thrives in ambiguity.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, Java, SQL

Frontend: React.js (React 18), HTML5, CSS3, Tailwind CSS

Backend & APIs: Node.js, Express.js, REST API Design, FastAPI, Authentication (OAuth2, Passport, BCrypt)

Databases: SQL & Database Design (PostgreSQL, SQLite), MongoDB (Atlas), ChromaDB (Vector Database)

AI / LLM: LLM Integration (Google Gemini API), Embeddings, Vector Databases, RAG Pipelines, LangGraph, LangChain, Agentic & Tool-Calling Workflows, Prompt Engineering

Tools & Practices: Git, GitHub, GitHub Actions (CI/CD), Docker, pytest, Agile / SDLC

EXPERIENCE

Banyanbrain Digital Pvt. Ltd. — Software Development Engineer Intern *Remote | Dec 2025 – Feb 2026*

- Contributed to full-stack feature development in an agile team over a 62-day internship, collaborating with senior developers on technical design.
- Wrote programs, created test data, and tested and debugged features end to end using Git.

1M1B (One Million for One Billion) — Data Analytics Intern *Remote | Jun 2026 – Sep 2026*

- Completed a competitive 4-month program in Python, Advanced Excel and Power BI.
- Built interactive dashboards and automation scripts that turned raw data into actionable, client-facing reports.

PROJECTS

JARVIS-Style AI Desktop Assistant for Windows ***IN PROGRESS***

Python 3.12, asyncio, PySide6, QML, Qt Quick 3D, Pydantic, SQLite (FTS5), faster-whisper, pytest, Astra 6 (AI-assisted development)

- Architecting a privacy-first AI assistant: a QML desktop interface (conversation, optional 3D avatar, approvals) driving an asyncio-based local orchestrator for session context, task state and cancellation.
- Designing 3 modular services (voice with local faster-whisper STT and Windows SAPI TTS, SQLite + FTS5 memory, and vision) behind typed events, with a cloud LLM adapter that receives approved context only.
- Building a local policy broker that validates every LLM-proposed tool call against Pydantic schemas through a 5-stage pipeline (schema → scope → risk → disclosure → human confirmation) before a sandboxed executor runs file, app, system, browser, Git and approved-command operations with timeouts, verification and redacted audit logs.

Doc-Aware Research Agent — Agentic RAG Assistant Live Demo

Python, LangGraph, Google Gemini, ChromaDB, Embeddings, Tavily / DuckDuckGo, RAG

- Built a LangGraph agent that routes each question to the user's uploaded documents first (ChromaDB vector search over embeddings) and falls back to live web search, telling the user which source answered.
- Developed the ingestion pipeline for 4 file formats (PDF, TXT, MD, DOCX) with chunking, embedding and instant indexing with Gemini generating grounded responses; deployed on Render.

ELITE MOTORS — Full-Stack Vehicle Showcase GitHub | Live Demo

React 18, TypeScript, Vite, Node.js, Express, PostgreSQL (SQL), Drizzle ORM, Zod, Tailwind CSS, Framer Motion, Replit Agent

- Built and deployed a full-stack React/TypeScript + Node.js/Express application with a PostgreSQL schema (Drizzle ORM, Zod validation) spanning 3 vehicle categories (cars, bikes, private jets) including an Indian-market collection.
- Implemented session-based authentication (Passport, OpenID Connect), REST endpoints, an admin CRUD dashboard, live search and a favorites system; added a drag-to-rotate 360° viewer and Framer Motion UI.

PawFind — Pet Adoption Portal GitHub | Live Demo

Java, Spring Boot 3, MongoDB Atlas, Spring Security, Thymeleaf, Three.js, Docker

- Built and deployed a REST application (Controller–Service–Repository) on MongoDB Atlas with search, wishlist, a 3-stage status-tracked adoption workflow (Submitted → Under Review → Approved/Rejected) and an admin dashboard.
- Secured with Spring Security (BCrypt, Google OAuth2, CSRF protection, role-based USER/ADMIN access); containerized with Docker.

SWIFT-SPLIT 2.0 — Secure File Fragmentation System GitHub | Live Demo

Python, FastAPI, AES-256, Reed-Solomon Erasure Coding, Shamir's Secret Sharing, Three.js

- Built a FastAPI service that splits files into k-of-n encrypted shards and reconstructs them byte-for-byte from any k shards; wrote AWS Signature V4 and WebDAV clients from scratch, validated against AWS test vectors and an 8-test suite.

Additional: CEREVIA 2.0 (GitHub) — C++/Spring Boot/Flask microservices behind a Gemini 2.0 LLM chatbot with VADER sentiment and crisis detection; SocialPulse V2 (GitHub) — Flask forecasting pipeline (Weibull + RK4), 1.2%–8.1% median error on a 180-video backtest, Docker + pytest; MY-PORTFOLIO (GitHub) — Spring Boot API + Three.js portfolio with CI/CD via GitHub Actions; Sorting Visualizer, SQL Query Xeno, LeetCode DSA Vault.

EDUCATION

Chandigarh University — B.E. Computer Science Engineering | Currently in 7th Semester *Aug 2023 – May 2027*

CERTIFICATIONS & ACHIEVEMENTS

- Wipro Certified: Java Full Stack | Anthropic: Claude Code 101 (Sep 2026) | Samsung Innovation Campus AI Certification | Coursera: Java | 2 published research papers | 2nd Rank, Maths Carnival